

Diagram illustrating a graph neural network architecture for handwritten digit recognition. The input image of the digit '7' is processed by an encoder (E) to produce a node embedding (blue dot). This embedding is part of a graph structure with other nodes (grey and pink dots) and edges. A decoder (D) processes the graph to produce a reconstruction (yellow dot). The reconstruction is compared with the input image using a loss function L , which is used to calculate the gradient of the loss with respect to the node embedding, denoted as $\Delta g_i = -\eta e_i \delta_n$. The diagram also shows a reconstruction path z and a reconstruction error δ_0 .

The diagram illustrates a hierarchical generative model for a 12x12 grid. At the bottom, a yellow node labeled δ_0 is connected to three gray nodes. Each gray node is connected to a group of four white nodes. To the right, a vertical stack of three gray rectangles represents the grid, with labels $K=1$, $K=3$, and $K=12$ indicating different levels of detail.

Figure 1 is a scatter plot with box plots showing the performance of Shunting (green) and Additive (blue) models across four datasets: MNIST fresh, MNIST DFA, Fashion-MNIST, and CIFAR-10. The x-axis represents the performance gain per neuron relative to a scalar baseline, measured in 'pp' (bits per neuron), ranging from 0 to 25. A vertical dashed line is at x=0. For each dataset, there are two box plots: one for Shunting (green) and one for Additive (blue). The Shunting models generally show higher performance gains than Additive models across all datasets, with the most significant gains observed in the CIFAR-10 dataset.

The diagram illustrates three different ways to compute the output of a neural network, each shown in a separate box with a title and a corresponding diagram.

- Strict scalar:** The diagram shows a network with multiple input nodes at the top, each receiving a scalar input (represented by orange arrows). These inputs are processed through several layers of nodes (represented by circles) to produce a single output node (yellow circle) labeled δ_0 . The title "Strict scalar" is in a light blue box.
- Per neuron:** The diagram shows a network with multiple input nodes at the top, each receiving a scalar input (represented by orange arrows). These inputs are processed through several layers of nodes (represented by circles) to produce a single output node (yellow circle) labeled δ_0 . A green box labeled "local rule" is positioned above the output node. The title "Per neuron" is in a light green box.
- Exact path:** The diagram shows a network with multiple input nodes at the top, each receiving a scalar input (represented by orange arrows). These inputs are processed through several layers of nodes (represented by circles) to produce a single output node (yellow circle) labeled δ_0 . The path from the input nodes to the output node is highlighted in red, with weights α_1 , α_2 , and α_3 labeled along the path. The title "Exact path" is in a light red box.

Below the "Exact path" diagram, the formula $\hat{\delta}_n = \alpha_3 \alpha_2 \alpha_1 \delta_0$ is shown.

Test accuracy (%)

Strict scalar Per neuron Projected K = 1 Projected K = 3 Exact path Decoder only

Additive

Shunting

97.4
97.2
97.0

Figure 1 is a dot plot showing the distribution of exact paths per neuron (pp) for different datasets and methods. The x-axis ranges from -3.0 to 1.0. The y-axis lists datasets and methods: MNIST fresh, MNIST DFA, Fashion-MNIST, CIFAR-10, and CIFAR-10 exact - BP. Data points are green circles (fresh) and blue squares (DFA). A vertical dashed line is at 0.0. A shaded gray box highlights the range from approximately -1.0 to 0.5 for the CIFAR-10 exact - BP method.

Figure 10 is a line graph showing the mean capture fraction for two models, Additive and Shunting, across different numbers of profiles (K). The Y-axis represents the mean capture fraction, ranging from 0.50 to 1.00. The X-axis represents the number of profiles (K), with values 1 and 3. A dashed horizontal line at Y=1.00 is labeled K=12. The Additive model (blue line) shows a mean capture fraction of approximately 0.78 at K=1 and increases to approximately 0.88 at K=3. The Shunting model (green line) shows a mean capture fraction of approximately 0.62 at K=1 and increases to approximately 0.65 at K=3. Both models show improvement as the number of profiles increases. The legend indicates that open circles represent the initial state and filled circles represent the trained state.

Model	Profiles K	Initial (Open Circle)	Trained (Filled Circle)
Additive	1	~0.78	~0.78
	3	~0.88	~0.88
Shunting	1	~0.62	~0.62
	3	~0.65	~0.65